| Proudcutions (Rules to be implemented in Phase 1 Submission are highlighted) |
|---|
| |
| start : program ; |
| |
| program |
| : program unit |
| \| unit |
| ; |
| |
| unit |
| : var_declaration |
| \| func_declaration |
| \| func_definition |
| ; |
| |
| func_declaration |
| : type_specifier ID LPAREN parameter_list RPAREN SEMICOLON |
| \| type_specifier ID LPAREN RPAREN SEMICOLON |
| ; |
| |
| func_definition |
| : type_specifier ID LPAREN parameter_list RPAREN compound_statement |
| \| type_specifier ID LPAREN RPAREN compound_statement |
| ; |
| |
| parameter_list |
| : parameter_list COMMA type_specifier ID |
| \| parameter_list COMMA type_specifier |
| \| type_specifier ID |
| \| type_specifier |
| ; |
| |
| compound_statement |
| : LCURL statements RCURL |
| \| LCURL RCURL |
| ; |
| |
| var_declaration |

| |
|---|
| : type_specifier declaration_list SEMICOLON ; |
| |
| type_specifier |
| : INT |
| \| FLOAT |
| \| VOID |
| ; |
| |
| declaration_list |
| : declaration_list COMMA ID |
| \| declaration_list COMMA ID LTHIRD CONST_INT RTHIRD |
| \| ID |
| \| ID LTHIRD CONST_INT RTHIRD |
| ; |
| |
| statements |
| : statement |
| \| statements statement |
| ; |
| |
| statement |
| : var_declaration |
| \| expression_statement |
| \| compound_statement |
| \| FOR LPAREN expression_statement expression_statement expression RPAREN statement |
| \| IF LPAREN expression RPAREN statement |
| \| IF LPAREN expression RPAREN statement ELSE statement |
| \| WHILE LPAREN expression RPAREN statement |
| \| PRINTLN LPAREN ID RPAREN SEMICOLON |
| \| RETURN expression SEMICOLON |
| ; |
| |
| expression_statement |
| : SEMICOLON |
| \| expression SEMICOLON |
| ; |
| |
| variable |

| |
|---|
| : ID |
| \| ID LTHIRD expression RTHIRD |
| ; |
| |
| expression |
| : logic_expression |
| \| variable ASSIGNOP logic_expression |
| ; |
| |
| logic_expression |
| : rel_expression |
| \| rel_expression LOGICOP rel_expression |
| ; |
| |
| rel_expression |
| : simple_expression |
| \| simple_expression RELOP simple_expression |
| ; |
| |
| simple_expression |
| : term |
| \| simple_expression ADDOP term |
| ; |
| |
| term |
| : unary_expression |
| \| term MULOP unary_expression |
| ; |
| |
| unary_expression |
| : ADDOP unary_expression |
| \| NOT unary_expression |
| \| factor |
| ; |
| |
| factor |
| : variable |
| \| ID LPAREN argument_list RPAREN |

| |
|---|
| \| LPAREN expression RPAREN |
| \| CONST_INT |
| \| CONST_FLOAT |
| \| variable INCOP |
| \| variable DECOP |
| ; |
| |
| argument_list |
| : arguments |
| \| |
| ; |
| |
| arguments |
| : arguments COMMA logic_expression |
| \| logic_expression |
| ; |